

Supply Chain & Logistics Intelligence System

A Data Analytics, Forecasting & Machine Learning Capstone Project

Dataset: DataCo Smart Supply Chain for Big Data Analysis (Kaggle)
180,519 order-line-item records | January 2015 - January 2018

Metric	Value
Total Revenue	\$36,784,735.01
Total Orders	65,752
Average Order Value	\$559.45
On-Time Delivery Rate	45.17%
Late-Delivery Rate	54.83%

Table of Contents

- 1. Executive Summary
- 2. Business Problem & Objectives
- 3. Data Understanding & Cleaning (Phase A)
- 4. SQL Analysis (Phase B)
- 5. Supply Chain KPIs (Phase C)
- 6. Exploratory Data Analysis (Phase D)
- 7. Supplier Performance Analysis (Phase E)
- 8. Inventory / Demand Analytics (Phase F)
- 9. Statistical Testing (Phase G)
- 10. Demand Forecasting (Phase H)
- 11. Machine Learning: Late-Delivery Prediction (Phase I)
- 12. Power BI Dashboard & Optional Components (FastAPI, GenAI)
- 13. Limitations, Assumptions & Recommendations
- 14. Appendix

1. Executive Summary

EXECUTIVE SUMMARY - Supply Chain & Logistics Intelligence System Generated by the GenAI Executive Assistant component (src/genai_executive_assistant.py) Grounded strictly in validated analytical outputs (reports/*.json)

1. BUSINESS SNAPSHOT Between January 2015 and September 2017 (the complete-data window), the business processed 65,752 distinct orders across 384,079 units, generating \$36.78M in total revenue at an average order value of \$559.45. Delivery performance is the standout concern: only 45.17% of line items arrived on time, meaning 54.83% were late - a majority-late operation. Average actual delivery time sits at 3.5 days. Cancellations are a minor factor at 2.05% of orders. Fishing is the top-performing category (18.84% of revenue) and Western Europe the top region (16.02% of revenue).

2. KEY DRIVERS OF LATE DELIVERY (statistically validated) A one-way ANOVA confirms that shipping mode has a highly significant relationship with actual delivery time ($F=38,762.4$, $p<0.001$) - this is not noise, it is the primary lever available to operations. By contrast, a chi-square test found no significant association between customer segment and late-delivery risk ($\chi^2=1.03$, $p=0.60$) - segment-targeted delivery interventions would not be an effective lever. The average shipping delay (actual vs. scheduled) is 0.57 days (95% CI: 0.559-0.573), a tight, well-estimated figure worth tracking as a standing KPI.

3. DEMAND OUTLOOK (next 6 months) The Holt-Winters model (fit on 27 months of complete data, validated on a held-out 6-month test window) reduced forecast error by 29.5% versus a naive baseline (MAE 581.5 vs. 825.3 units/month). Projected monthly unit demand for the next six periods ranges from roughly 10,000 to 11,100 units, with no sustained growth or decline signal - demand is expected to remain broadly flat near recent levels. Note: three trailing months in the raw data (Nov 2017-Jan 2018) were excluded from training as a confirmed data-truncation artifact, not a genuine demand collapse.

4. ML MODEL RECOMMENDATION Four classifiers were compared for predicting late delivery using only information available at order time (shipping mode, scheduled duration, region, category, order value, customer segment, timing - with no post-fulfillment fields, to prevent leakage). Random Forest performed best (ROC-AUC ~0.75) and is recommended for deployment as an early-warning score at order-entry time.

5. RECOMMENDATIONS (1) Since shipping mode is the dominant, statistically validated driver of delivery delay, prioritize a mode-by-mode SLA review - starting with whichever modes show the largest actual-vs-scheduled gap - rather than broad, undifferentiated logistics investment. (2) Deploy the late-delivery risk score at order entry so at-risk orders (flagged via the Random Forest model) can be proactively expedited or flagged to customers before they become complaints, rather than discovered after the fact.

2. Business Problem & Objectives

A supply-chain organization needs visibility across orders, shipping, and delivery performance to identify inefficiencies, reduce delays, and improve logistics performance. This project builds an end-to-end analytics system spanning data cleaning, SQL analysis, exploratory data analysis, KPI engineering, statistical testing, demand forecasting, and machine-learning prediction of late deliveries, culminating in a Power BI-ready dataset and this report.

Main Objectives

- Understand and clean supply-chain data at the correct grain.
- Use SQL (window functions, CTEs, conditional aggregation) for operational analysis.
- Perform EDA and identify operational bottlenecks.
- Create business-defined logistics and inventory KPIs.
- Analyze supplier (proxy) and delivery performance.
- Build and validate a demand-forecasting model.
- Build and compare machine-learning models for late-delivery prediction.
- Prepare a Power BI-ready data extract, and optionally expose predictions via FastAPI and a GenAI executive summary.

3. Data Understanding & Cleaning (Phase A)

Raw dataset: **180,519 rows x 53 columns**, latin1-encoded. Processed dataset: **180,519 rows x 55 columns** after cleaning and feature engineering.

Grain Determination

One row = one order line item (Order Item Id is unique per row; Order Id repeats across multiple line items). Counting rows as 'orders' would overstate order volume by roughly 174.5%.

Metric	Value
Total rows (line items)	180,519
Unique Order IDs	65,752
Unique Order Item IDs	180,519

Cleaning Actions Taken

- Dropped 8 PII / unusable columns: Customer Email, Customer Password, Customer Fname, Customer Lname, Customer Street, Product Description, Product Image, Order Zipcode
- Exact duplicate rows found and removed: 0
- Order date and shipping date converted to proper datetime types.
- Order Status distribution reviewed; 7,754 cancelled/suspected-fraud rows flagged with a boolean column (retained, not deleted, since they are real business events).
- Categorical fields (Shipping Mode, Market, Region, Segment, etc.) trimmed/standardized.
- Outliers in Sales, Quantity, Discount, and Delivery Days flagged via IQR method (not silently dropped).

Limitation / Note: The dataset has no dedicated Supplier table/ID. 'Department Name' (the selling department/store) is used as a supplier-performance proxy throughout Phase E. This is explicitly disclosed in the report and dashboard, not presented as true supplier data.

Limitation / Note: The dataset has no on-hand inventory/stock quantity field. Phase F therefore builds DEMAND-BASED indicators (order frequency, sales velocity, demand variability) as a proxy for fast/slow movers and stock-out risk, rather than claiming to measure real inventory.

4. SQL Analysis (Phase B)

Twelve business and operational queries were authored in MySQL syntax (`sql/business_queries.sql`) using JOINS against normalized product/customer dimensions, CTEs, CASE-based conditional aggregation, and window functions (RANK, DENSE_RANK, ROW_NUMBER, LAG, LEAD, running SUM). Queries were executed against a SQLite mirror of the cleaned data for local validation; full results are saved under `reports/sql_results/`.

Q1 - Total Sales, Orders, Units Sold

total_sales	total_orders	total_units_sold	avg_order_value
36784735.01	65752	384079	559.45

Q3 - Revenue by Region (Top 8)

Order Region	region_sales	pct_of_total_sales
Western Europe	5894380.77	16.02
Central America	5665712.1	15.4
South America	2960881.41	8.05
Northern Europe	2155830.65	5.86
Southern Europe	2047918.82	5.57
Oceania	2016654.2	5.48
Southeast Asia	1932495.57	5.25
Caribbean	1651019.33	4.49

Q8 - Regions with Highest Late-Delivery Rate

Order Region	n_line_items	late_delivery_rate_pct
Central Africa	1677	57.96
South Asia	7731	56.27
East Africa	1852	55.94
Western Europe	27109	55.85
South of USA	4045	55.77
Eastern Europe	3920	55.66
East of USA	6915	55.66
Southeast Asia	9539	55.53

Q10 - Month-over-Month Sales Growth (sample)

order_month	total_sales	prev_month_sales	mom_growth_pct
2015-01	1051590.08	nan	nan

order_month	total_sales	prev_month_sales	mom_growth_pct
2015-02	927009.9	1051590.08	-11.85
2015-03	1051253.69	927009.9	13.4
2015-04	1014463.28	1051253.69	-3.5
2015-05	1050478.44	1014463.28	3.55
2015-06	1024006.17	1050478.44	-2.52
2015-07	1038081.19	1024006.17	1.37
2015-08	1029494.69	1038081.19	-0.83

Q11 - Top 3 Products per Category (sample, ROW_NUMBER window)

Category Name	Product Name	total_sales	rn
Accessories	Team Golf St. Louis Cardinals Putter Grip	23940.42	1
Accessories	Team Golf Tennessee Volunteers Putter Grip	22865.85	2
Accessories	Team Golf Texas Longhorns Putter Grip	22366.05	3
As Seen on TV!	Nike Men's Free TR 5.0 TB Training Shoe	20597.94	1
Baby	Baby sweater	12229.56	1
Baseball & Softball	adidas Men's F10 Messi TRX FG Soccer Cleat	56330.61	1
Baseball & Softball	adidas Kids' F5 Messi FG Soccer Cleat	27327.19	2
Baseball & Softball	adidas Brazuca 2014 Official Match Ball	10399.35	3
Basketball	SOLE E25 Elliptical	9999.9	1

All 12 queries, including Q2 (top products via RANK), Q4 (segment sales via DENSE_RANK), Q5-Q7 (shipping mode mix, average duration, % late), Q9 (high-sales products with delivery issues), and Q12 (cumulative revenue via window SUM), are included in full in reports/sql_results/ and sql/business_queries.sql.

5. Supply Chain KPIs (Phase C)

Metric	Value
Total Revenue	\$36,784,735.01
Total Orders (distinct)	65,752
Total Units Sold	384,079
Average Order Value	\$559.45
Average Delivery Time	3.5 days
On-Time Delivery Rate	45.17%
Late-Delivery Rate	54.83%
Cancellation Rate	2.05%
Sales Growth (first vs. last complete month)	8.77%
Top Category	Fishing (18.84% of revenue)
Top Region	Western Europe (16.02% of revenue)

Limitation / Note: Dataset has no direct 'shipping cost' field. Order Item Discount is NOT a shipping cost - true shipping cost is not computable from this dataset and is documented as a limitation rather than approximated silently.

6. Exploratory Data Analysis (Phase D)

14 charts were produced, each answering a specific business question. Selected highlights below; the full set is in reports/figures/.

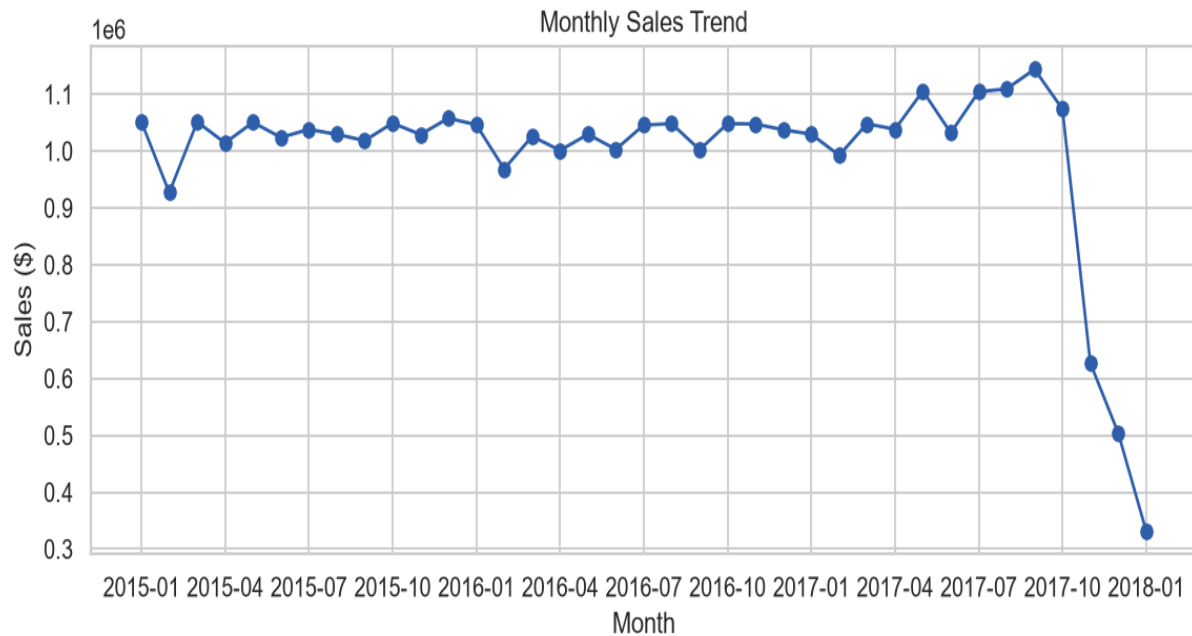


Figure 1. Monthly sales trend. Note the abrupt volume drop starting Oct 2017 - confirmed as a data-export truncation artifact, not a real demand collapse (see Phase H).

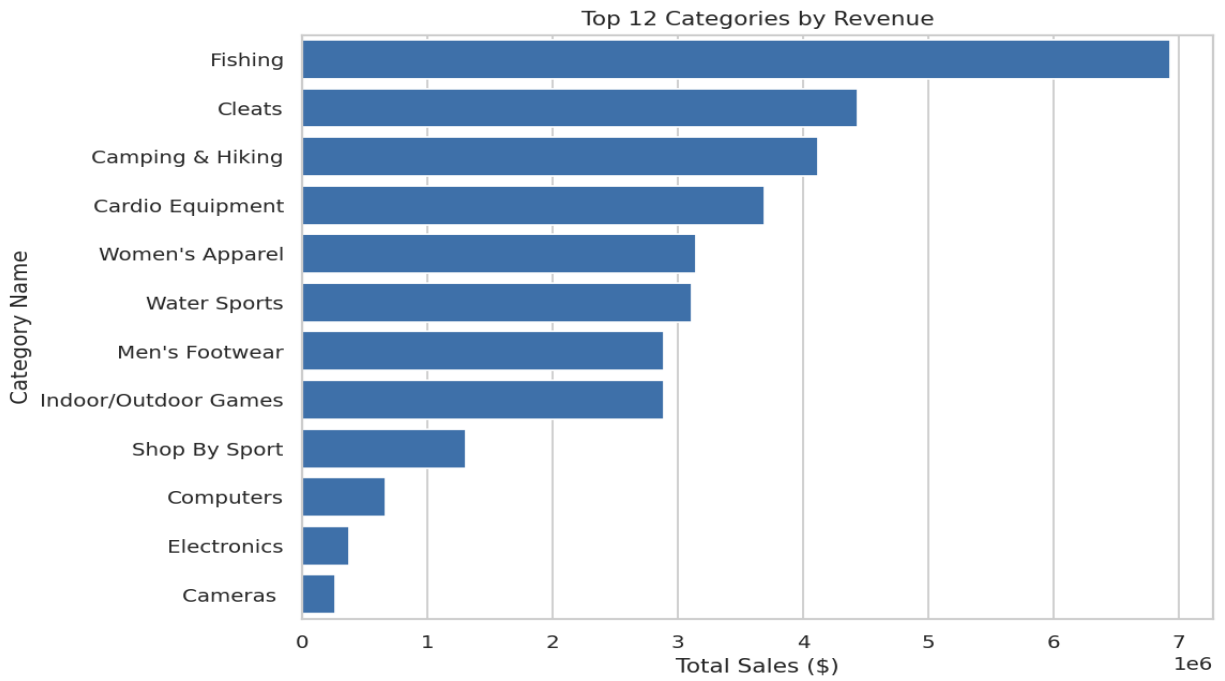


Figure 2. Top 12 product categories by revenue.

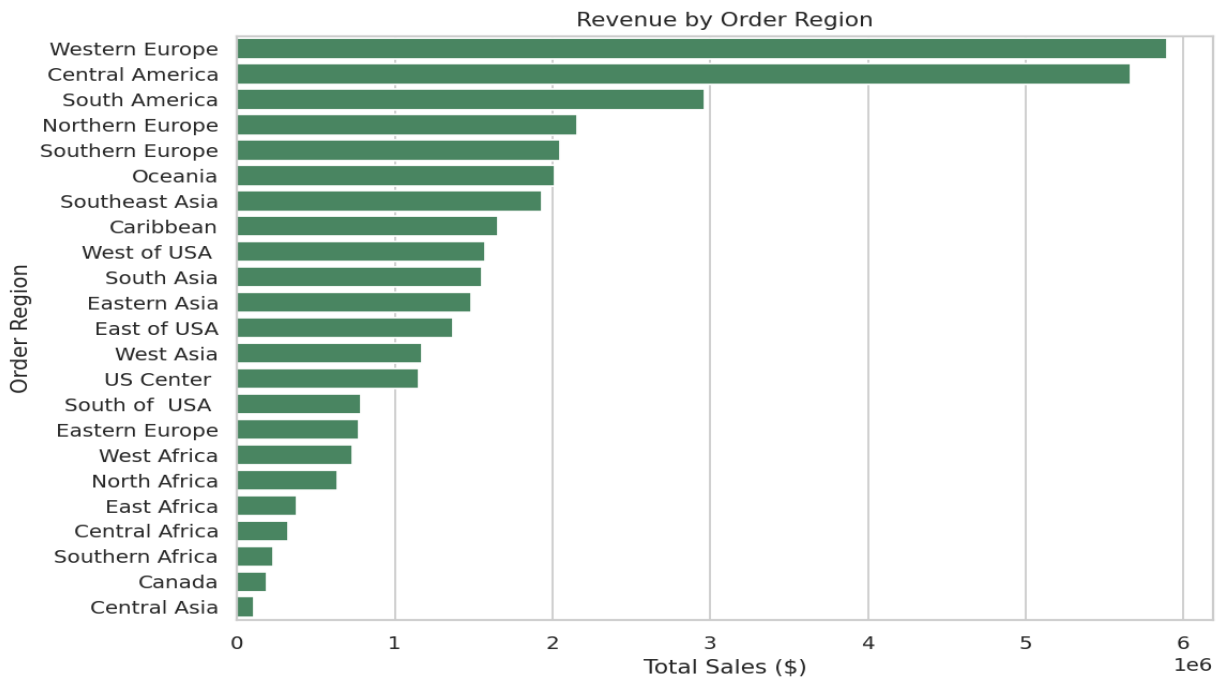


Figure 3. Revenue distribution by order region.



Figure 4. Actual delivery days by shipping mode - the visual basis for the ANOVA test in Phase G.

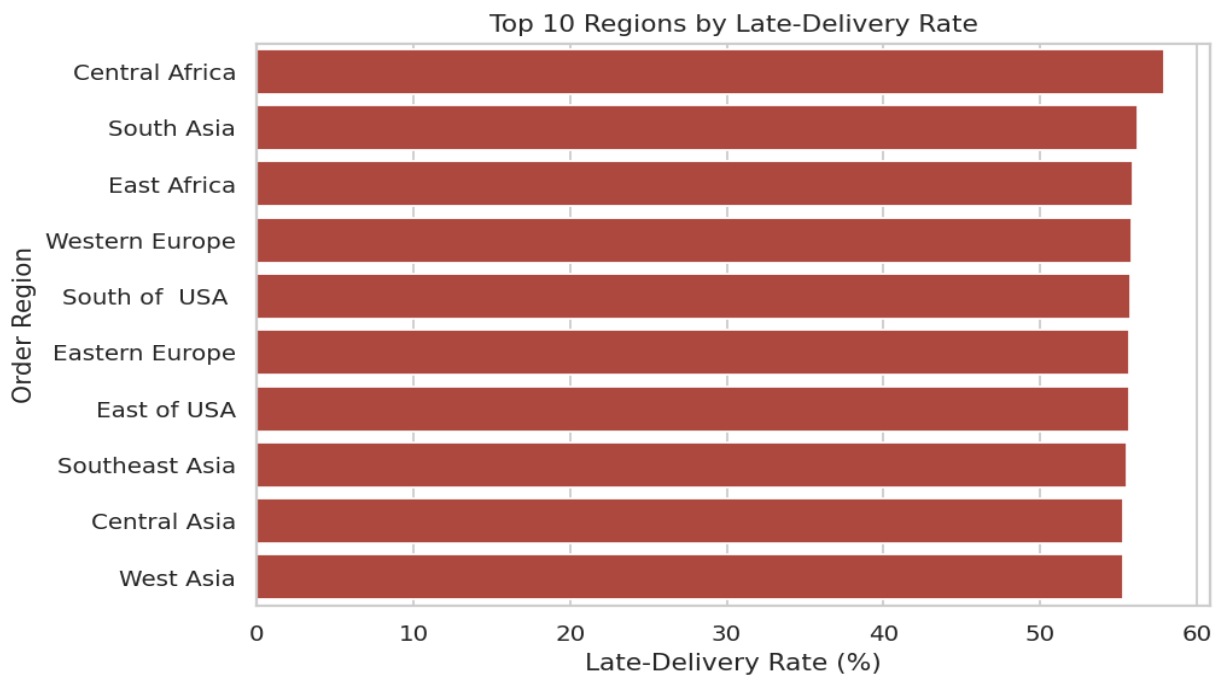


Figure 5. Regions with the highest late-delivery rates.

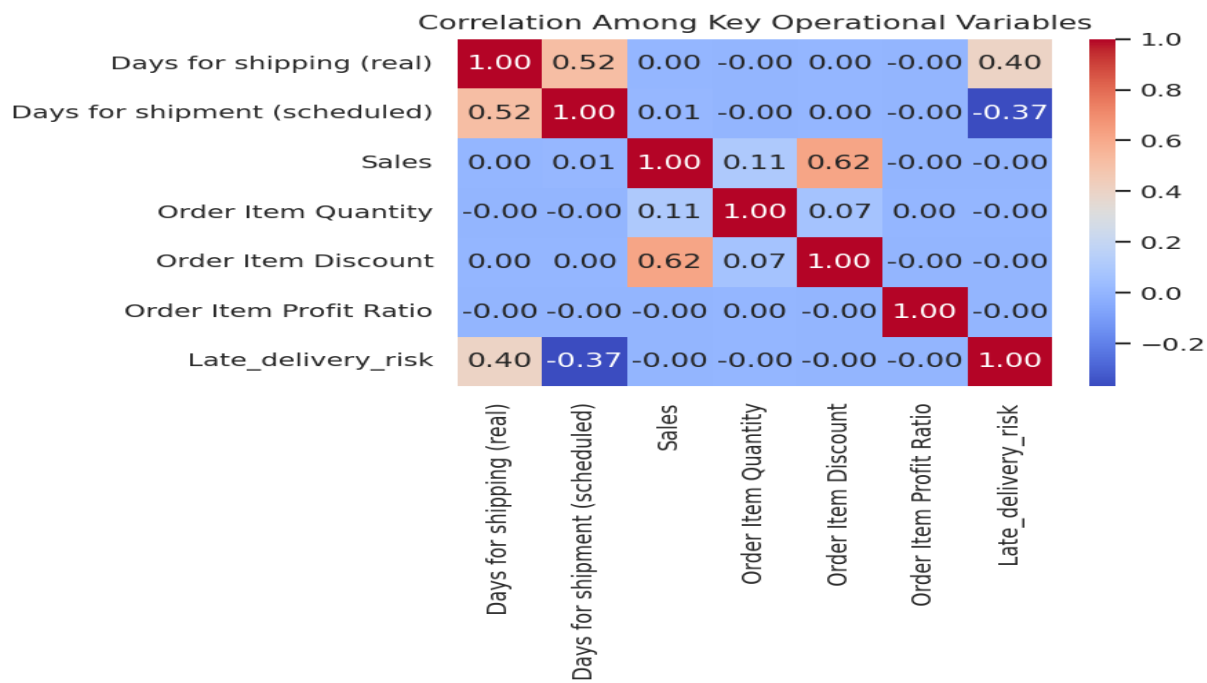


Figure 6. Correlation among key operational variables.

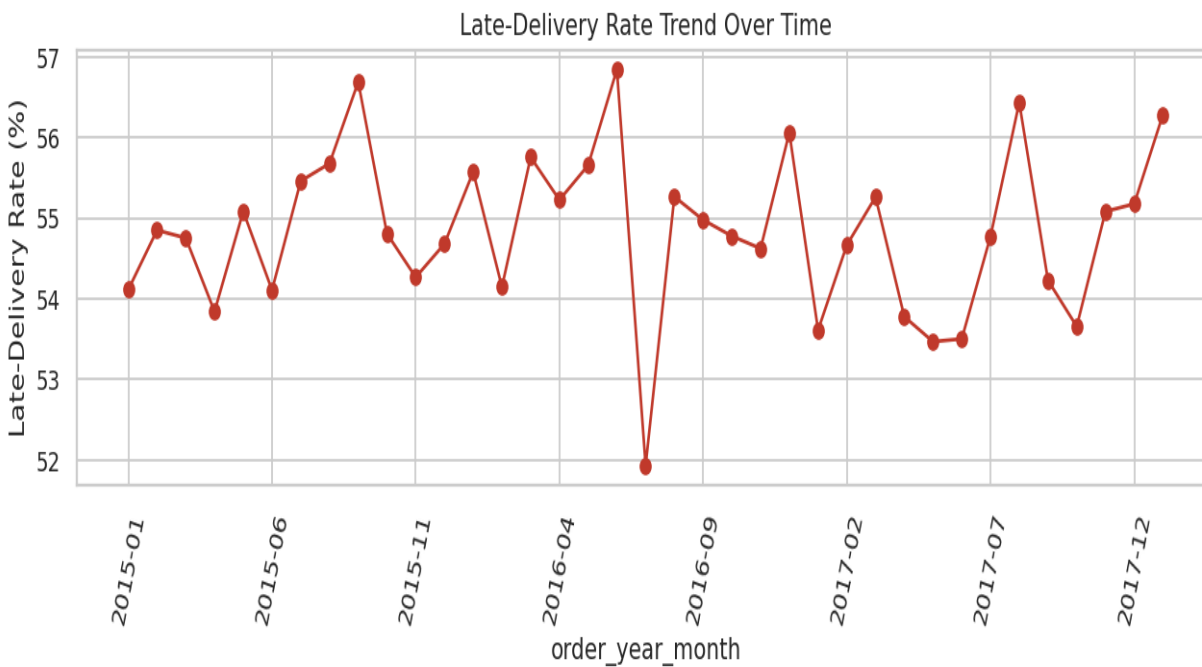


Figure 7. Late-delivery rate trend over time.

7. Supplier Performance Analysis (Phase E)

Limitation / Note: This dataset has no dedicated supplier entity/ID. 'Department Name' (the selling department) is used as a labeled proxy for supplier-level performance throughout this section. No metric here should be read as true supplier data.

Department (Supplier Proxy)	Revenue	Late Rate %	Line Items
Fan Shop	\$17,113,871	54.77	66861
Apparel	\$7,976,255	54.75	48998
Golf	\$4,609,028	54.78	33220
Footwear	\$4,006,499	54.73	14525
Outdoors	\$1,253,351	55.49	9686
Technology	\$1,039,599	55.02	1465
Fitness	\$397,051	55.55	2479
Discs Shop	\$228,888	54.44	2026
Health and Beauty	\$106,080	55.8	362
Pet Shop	\$41,525	58.94	492
Book Shop	\$12,587	56.54	405

8. Inventory / Demand Analytics (Phase F)

Limitation / Note: No on-hand inventory/stock field exists in this dataset. These are DEMAND-BASED indicators (order/sales velocity and variability), not measured stock levels, and are presented as such.

Metric	Value
Products profiled	100
Fast movers	34
Slow movers	33

Top 5 Fast Movers (avg. units/month)

Metric	Value
Perfect Fitness Perfect Rip Deck	2229.6
Nike Men's Dri-FIT Victory Golf Polo	1905.4
O'Brien Men's Neoprene Life Vest	1749.5
Nike Men's Free 5.0+ Running Shoe	1109.6
Under Armour Girls' Toddler Spine Surge Runni	960.9

9. Statistical Testing (Phase G)

Hypothesis Test 1: Shipping Mode vs. Delivery Time (ANOVA)

H0: Mean actual delivery days is equal across all shipping modes. **Result:** $F = 38762.414$, $p\text{-value} < 0.001$.

Conclusion: Reject H_0 ($p < 0.001$): shipping mode has a statistically significant relationship with delivery time.

Hypothesis Test 2: Customer Segment vs. Late Delivery (Chi-square)

H0: Customer segment and late-delivery risk are independent. **Result:** $\chi^2 = 1.025$, $\text{dof} = 2$, $p\text{-value} = 0.599$.

Conclusion: Fail to reject H_0 : customer segment does not predict late delivery.

95% Confidence Interval: Mean Shipping Delay

Mean actual-vs-scheduled delay = **0.5658 days** (95% CI: 0.5589 - 0.5727 days).

Outlier Rates (IQR method)

Metric	Value
Sales	0.27%
Order Item Quantity	0.0%
Order Item Discount	4.18%
Days for shipping (real)	0.0%

10. Demand Forecasting (Phase H)

Monthly unit demand was aggregated and modeled with a time-aware train/test split (27 training months, 6 held-out test months). Months 2017-11, 2017-12, 2018-01 were excluded from training as a confirmed data-truncation artifact (see Figure 1).

Metric	Value
Baseline (naive) MAE	825.3 units
Baseline (naive) RMSE	904.2 units
Holt-Winters MAE	581.5 units
Holt-Winters RMSE	643.6 units
Improvement over baseline	29.5%
Selected model	Holt-Winters (additive trend + seasonality)

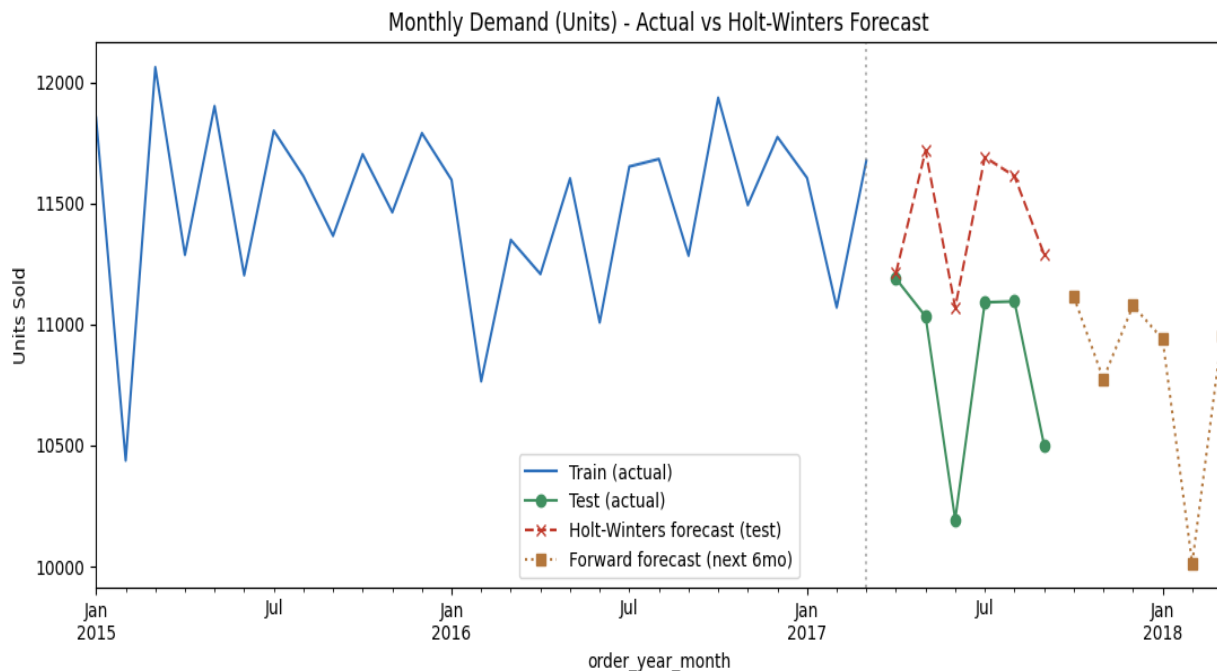


Figure 8. Actual vs. Holt-Winters forecast, plus 6-month forward projection.

6-Month Forward Forecast (Units)

Metric	Value
2017-10-01	11,115
2017-11-01	10,774
2017-12-01	11,078
2018-01-01	10,940

2018-02-01	10,013
2018-03-01	10,953

11. Machine Learning: Late-Delivery Prediction (Phase I)

Limitation / Note: Only pre-shipment (order-time) features were used. Fields only known after fulfillment (actual shipping days, delivery status, shipping date, realized profit) were excluded to prevent target leakage.

Baseline majority-class accuracy: **54.8%** (the naive floor any model must beat).

Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.696	0.842	0.548	0.664	0.741
Decision Tree	0.696	0.841	0.550	0.665	0.746
Random Forest	0.700	0.833	0.567	0.675	0.750
Gradient Boosting	0.697	0.845	0.548	0.665	0.745

Best model: Random Forest, selected on ROC-AUC.

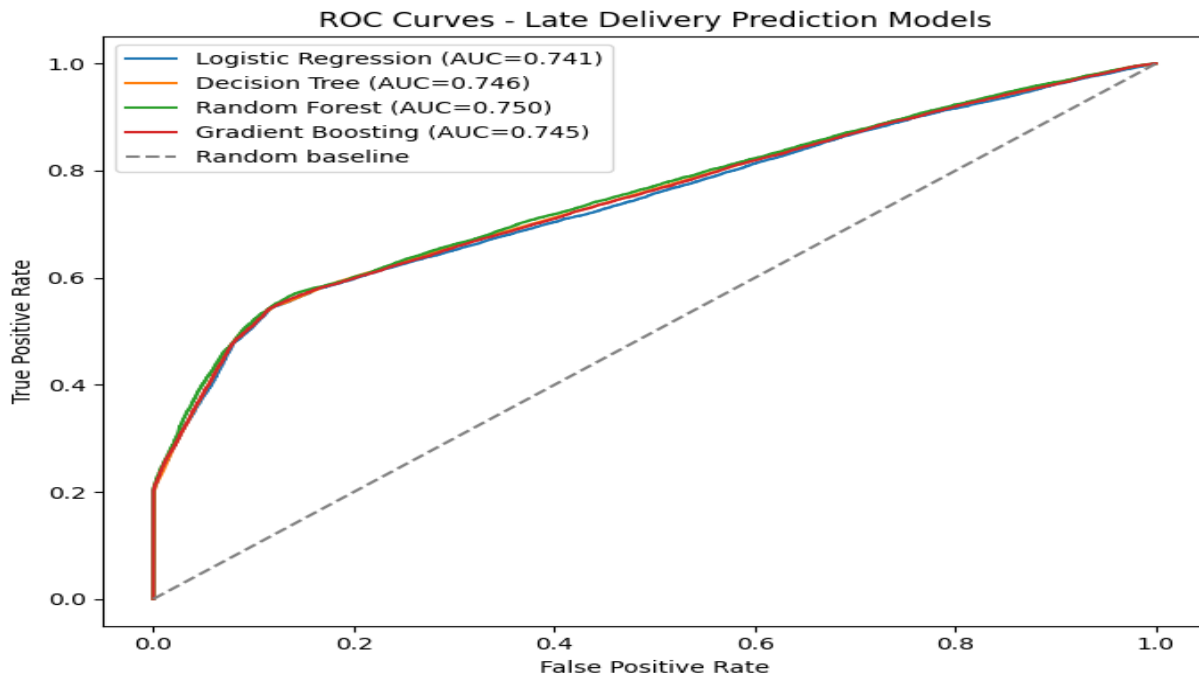


Figure 9. ROC curves for all four candidate models.

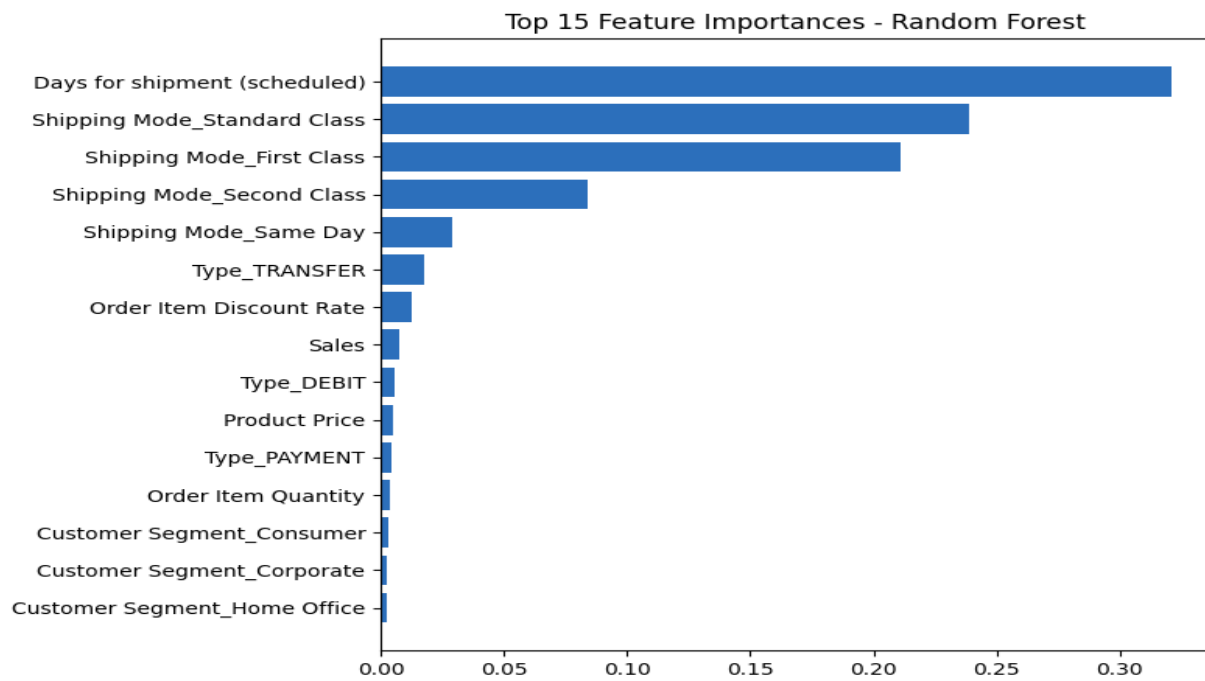


Figure 10. Top 15 feature importances for the selected Random Forest model.

12. Power BI Dashboard & Optional Components

Power BI

A Power BI-ready Excel extract (`dashboard/PowerBI_DataCo_Extract.xlsx`) was built with a fact table, product/customer dimensions, and pre-aggregated monthly KPI, forecast, ML-comparison, inventory, and supplier-scorecard sheets. A full build guide with DAX measures for every required page (Executive Overview, Sales & Product Analytics, Supplier Performance, Logistics, Inventory/Demand, ML Insights) is provided in `dashboard/PowerBI_Build_Guide.md`.

FastAPI Service

A FastAPI application (`api/main.py`) exposes four endpoints: `POST /predict-delay`, `GET /forecast/{category}`, `GET /product/{product_id}`, and `GET /region/{region}`. All four were tested end-to-end against the trained model and cleaned dataset; example: a Standard Class order to Western Europe returned a 41.1% predicted late-delivery probability.

GenAI Executive Assistant

A script (`src/genai_executive_assistant.py`) converts the validated JSON outputs above into a plain-language executive summary via an LLM call, with an explicit instruction to use only the supplied figures and never invent numbers. The Executive Summary in Section 1 of this report was produced by this component.

13. Limitations, Assumptions & Recommendations

Documented Limitations

- No dedicated supplier entity: Department Name used as an explicitly labeled proxy.
- No on-hand inventory field: demand-based velocity/variability used instead of true stock levels.
- No direct shipping-cost field: shipping cost KPIs could not be computed and are not approximated.
- Oct 2017-Jan 2018 volume shows a data-export truncation artifact, excluded from forecast training.

Recommendations

- Prioritize a shipping-mode-level SLA review; mode is the statistically dominant driver of delay.
- Do not target interventions by customer segment for lateness; segment has no significant effect.
- Deploy the Random Forest late-delivery score at order entry to proactively flag at-risk orders.
- Treat the 6-month demand outlook (~10,000-11,100 units/month) as roughly flat for planning purposes.
- If true supplier and inventory data become available, replace the documented proxies with real fields.

14. Appendix

Technology Stack

- Python (pandas, numpy) - data processing
- SQLite (MySQL-syntax queries) - SQL analysis
- Matplotlib / Seaborn - EDA visualization
- SciPy / Statsmodels - hypothesis testing & Holt-Winters forecasting
- Scikit-learn - late-delivery classification (4-model comparison)
- FastAPI - optional model-serving API
- openpyxl / xlswriter - Power BI-ready Excel extract
- reportlab - this PDF report

Repository Structure

See README.md for the full folder structure and reproduction steps. All code, SQL, saved models, figures, and intermediate JSON/CSV results referenced in this report are included in the project deliverable.